

Doubts Session 1

01. Prefix sum in matrix ✓
02. Next permutation ✓
03. No. of digit one (Maths)
04. Strange Equality ✓
05. K^* symbol → Easy & hard ✓
06. Letter phone ✓
07. subsets I → Subsets II
- * Subarray OR
- * Rain water Trapped

— ^ — ^ — ^ — ^ — ^ — ^ — ^

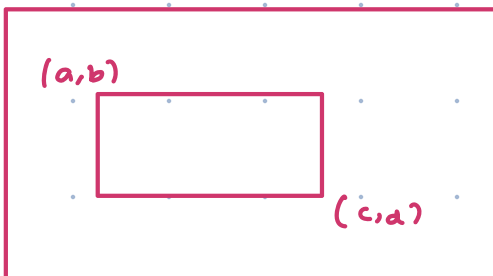
Submatrix → Part of a matrix

To represent a

submatrix → we don't need 4 corners

→ we just need TL & BR

→ TR & BL



Q Given a matrix of size $N \times M$ & Q queries
For each query, find sum of given submatrix

Note :- TL is top left & BR is bottom right.

	0	1	2	3
0	2	-1	3	2
1	3	2	6	2
2	10	9	8	2
3	4	-1	2	3
4	3	2	6	9

Queries = 2

(T,L) (B,R)

01. (2,1) & (4,2) = 26

02. (1,1) & (3,3) = 33

Brute force $\rightarrow$ For each query, iterate on the submatrix & get the sum.

TC: $O(Q \times m \times n)$

SC: $O(1)$

Idea 2 = Use psum technique

1D Array $\Rightarrow$ $\text{psum}[i]$ = sum of all elements from 0 to i

2D Array = $psum[i][j]$ $\Rightarrow$ sum of all elements from $(0,0)$ to (i,j)

$psum[4][4] = 2$

	0	1	2	3	4
0					
1					
2					
3					
4					

Annotations:

- $psum[0][3]$ (points to row 0, column 3)
- $psum[2][3]$ (points to row 2, column 3)
- $psum[3][2]$ (points to row 3, column 2)

* Assume psum matrix

	0	1	2	3	4	5	6
0							
1							
2							
3							
4							
5							
6							

Query

$(2,2) \rightarrow (5,4)$

$psum(5,4) - psum(5,1)$
 $- psum(1,4) + psum(1,1)$

	0	1	2	3	4
0					
1					
2					
3					
4					

$psum[][]$

Query

$(1, 2) \rightarrow (4, 3)$

$$psum(4, 3) - psum(4, 1) - p(0, 3) + psum(0, 1)$$

	0	1	$b_1 - 1$	b_1	4	b_2	6
0							
$a_1 - 1$							
a_1							
3							
a_2							
5							
6							

TL
 (a_1, b_1)

BR
 (a_2, b_2)

$$psum(a_2, b_2) - psum(a_2, b_1 - 1) - psum(a_1 - 1, b_2) + psum(a_1 - 1, b_1 - 1)$$

Query = a_1, b_1 to a_2, b_2

ans = psum(a_2, b_2)

if ($b_1 > 0$) { ans = ans - psum($a_2, b_1 - 1$)

if ($a_1 > 0$) { ans = ans - psum($a_1 - 1, b_2$)

if ($a_1 > 0$ & $b_1 > 0$) { ans = ans + psum($a_1 - 1, b_1 - 1$)

* Build the psum matrix

→ Apply psum in rows

→ Apply psum in cols

	0	1	2
0	a_0	b_0	c_0
1	a_1	b_1	c_1
2	a_2	b_2	c_2

arr



	0	1	2
0	a_0	$a_0 + b_0$	$a_0 + b_0 + c_0$
1	$a_0 + a_1$	$a_0 + b_0 + a_1 + b_1$	$a_0 + b_0 + c_0 + a_1 + b_1 + c_1$
2	$a_0 + a_1 + a_2$	$a_0 + b_0 + a_1 + b_1 + a_2 + b_2$	$a_0 + b_0 + c_0 + a_1 + b_1 + c_1 + a_2 + b_2 + c_2$

psum

	0	1	2	3
0	3	2	4	1
1	-1	4	3	2
2	2	7	6	3



	0	1	2	3
0	3	5	9	10
1	2	8	15	18
2	4	17	30	36

psum

	0	1	2	3
0	3	2	4	1
1	-1	4	3	2
2	2	7	6	3

Apply psum
in rows

	0	1	2	3
0	3	5	9	10
1	-1	3	6	8
2	2	9	15	18

Apply psum in
cols

	0	1	2	3
0	3	5	9	10
1	2	8	15	18
2	4	17	30	36

* Using psum Approach

$$TC = O(n * m + Q)$$

$$SC = O(n * m)$$

* Next permutation

$$A[] = \{1, 2, 3\} \longrightarrow \text{Ans} = \{1, 3, 2\}$$

$$\text{permutations} = \{1, 2, 3\}$$

$$\{1, 3, 2\}$$

$$\{2, 1, 3\}$$

$$\{2, 3, 1\}$$

$$\{3, 1, 2\}$$

$$\{3, 2, 1\}$$

$$\text{ex. } A[] = \{3, 1, 2\} \Rightarrow \text{Ans} = \{3, 2, 1\}$$

$$\text{ex. } A[] = \{3, 2, 1\} \Rightarrow \text{Ans} = \{1, 2, 3\}$$

BF Approach $\rightarrow$ Edge case $\rightarrow$ Check if the array is in descending order $\rightarrow$ reverse the array.

- $\rightarrow$ Generate all permutations in sorted manner
- $\rightarrow$ Iterate on all permutations & get the next greater permutation

$$\text{TC: } O(N! * N)$$

* Idea 2

$A[] = \{ 2, 1, 5, 4, 3 \}$

$A[] = \{ 2 \ 3 \ 5 \ 4 \ 1 \}$

$A[] = \{ 2 \ 3 \ 1 \ 4 \ 5 \}$

* $A[] = \{ 6 \ 5 \ 3 \ 7 \ 4 \ 2 \ 1 \}$

Ans = $\{ 6 \ 5 \ 4 \ 7 \ 3 \ 2 \ 1 \}$

Ans = $\{ 6 \ 5 \ 4 \ 1 \ 2 \ 3 \ 7 \}$

* Steps

01. Find first dip coming from RHS = x
02. Find the next greater ele of x in RHS region
03. Swap(x, y)
04. reverse($x+1$ to $n-1$);

* No. of digit One (Maths)

Given A , find & return the no. of digit 1 appearing in all non-negative integers $\leq A$.

$$A = 10$$

{ 1 2 3 4 5 6 7 8 9 10 } Ans = 2

$$A = 15$$

{ 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 }

$$\underline{\underline{\text{Ans} = 8}}$$

* Brute force solution $\Rightarrow$

01. Generate all the no. from 1 to A .
& look for every digit.

$$O(A * \log(A))$$

* Optimal Solution

One's place $\Rightarrow \{1, 10\} \Rightarrow 1$

$\{11, 20\} = 1$

$$\text{No. of 1 at Ones place} = \left\lfloor \frac{A}{10} \right\rfloor * 1 + (A \% 10 > 0 ? 1 : 0)$$

$$A = 70 \quad \text{no. of ones} = \frac{70}{10} = 7$$

$$A = 30 \quad \text{no. of 1} = \frac{30}{10} = 3$$

$$A = 33 \quad \text{no. of 1} = \frac{33}{10} = 3$$

Tenth place

$$[1 \quad 100] = 10 \quad \{10, 11, 12, 13, 14, 15, 16, 17, 18, 19\}$$

$$[101 \quad 200] = 10$$

$$\text{No. of 10 places} = \left\lfloor \frac{A}{100} \right\rfloor * 10 + \min(\max(A \% 100 - 9, 0), 10)$$

$$A = 100, \text{ No. of 1 at tenth place} = \left\lfloor \frac{100}{100} \right\rfloor * 10 = 10$$

$$A = 200, \text{ No. of 1s at tenth place} = \left\lfloor \frac{200}{100} \right\rfloor * 10 = 20$$

$$A = 518, \text{ No. of 1s at tenth place} = \underbrace{\left\lfloor \frac{518}{100} \right\rfloor * 10}_{+} + \underbrace{518 \% 100}_{18 - 9} = 50 + 9 = \underline{\underline{59}}$$

$$A = 504$$

$$\left\lfloor \frac{504}{100} \right\rfloor * 10 + 504 \% 100 - 9$$

50

50

+

4 - 9

negative should
be considered
as zero

$$A = 529$$

$$\left\lfloor \frac{529}{100} \right\rfloor * 10 + \min(\max(529 \% 100 - 9, 0), 10)$$

50

50

+

20, 0

```
int solve (A)
```

```
    int ans = 0
```

```
    for (i = 1; i <= A; i *= 10)
```

```
        int div = i * 10
```

```
        ans += (A / div) * i + min(max(A % div - i + 1, 0), i)
```

3

3

12:33 → 12:43 pm

Strange Equality

A

x = greatest no. smaller than A (floor)

y = smallest no. greater than A (ceil)

$$x \wedge A = x + A \quad \&\& (x < A)$$

$$y \wedge A = y + A \quad \&\& (y > A)$$

* Once we have x & $y \rightarrow$ return $x \wedge y$

	6	5	4	3	2	1	0	
A =	1	0	1	1	0	0	0	$= 2^6 + 2^4 + 2^3$

$\wedge$

x =	0	1	0	0	1	1	1	
-----	---	---	---	---	---	---	---	--

$x \wedge A$	=	1	1	1	1	1	1	$= \underline{x + A}$
								$2^6 + 2^4 + 2^3 +$
								$2^5 + 2^2 + 2^1 + 2^0$

	7	6	5	4	3	2	1	0	
A =	0	1	0	1	1	0	0	0	$= 2^6 + 2^4 + 2^3$

$\wedge$

y =	1	0	0	0	0	0	0	0	
-----	---	---	---	---	---	---	---	---	--

$A \wedge y$	=	1	1	0	1	1	0	0	$= \underline{A + y}$
--------------	---	---	---	---	---	---	---	---	-----------------------

$$2^6 + 2^4 + 2^3 + 2^1$$

$$x=0, y=0, z=0$$

```
while (A > 0) {
```

if $((A \& 1) == 0)$

$$x = x \mid (1 \leq i)$$
$$\underline{\ell} = \bar{\ell} + 1$$
$$A = (A \gg 1)$$
$$Y = (1 \leq i) \quad \checkmark$$

return x^y ;

K Symbol

Given N, K . Find the K^{th} ele in N^{th} row

$$0 \rightarrow \{0, 1\}$$
$$i \rightarrow \{1, 0\}$$
[illegible]

row
n

col
k

2	1	→	1
3	2	→	1
4	5	→	0

```
main ( ) {
```

```
    AL<I> ans = recur (n, k)
```

```
    return ans.get(k);
```

```
}
```

```
AL<I> recur (int n, int k) {
```

```
    if (n == 0) {
```

```
        AL<I> base = new AL<>();
```

```
        base.add(0);
```

```
        return base;
```

```
    }
```

```
    AL<I> rans = recur (n-1, k)
```

```
    AL<I> ans = new AL<>();
```

```
    for (i=0; i<rans.size(); i++) {
```

```
        int val = rans.get(i);
```

```
        if (val == 0) { ans.add(0);
```

```
                        ans.add(1);
```

```
        if (val == 1) { ans.add(1);
```

```
                        ans.add(0);
```

```
    }
```

```
    return ans;
```

```
}
```

$$TC: n \rightarrow (n-1)^1 \rightarrow (n-2)^1 \rightarrow (n-3)^1 \dots 1 \ 0$$

$$2^n + 2^{n-1} + 2^{n-2} + 2^{n-3} + \dots + 2^1 + 2^0$$

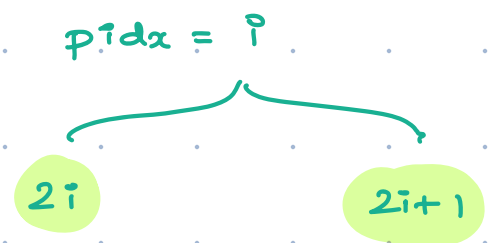
$$TC = O(2^n)$$

$$SC: O(2^n)$$

* Optimised solution

Observation $\rightarrow$ Every par is going to generate two children.

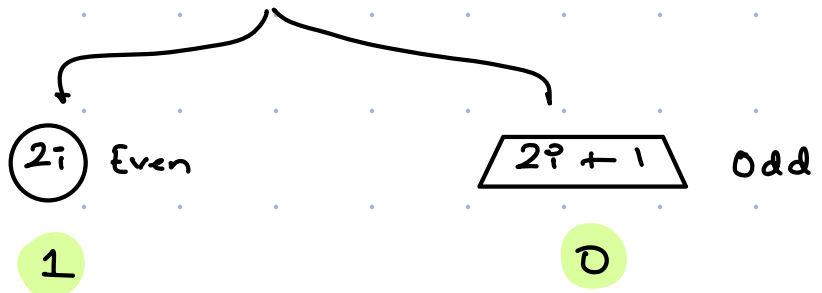
<u>idx</u>	<u>children idx</u>
0	0 & 1
1	2 & 3
2	4 & 5



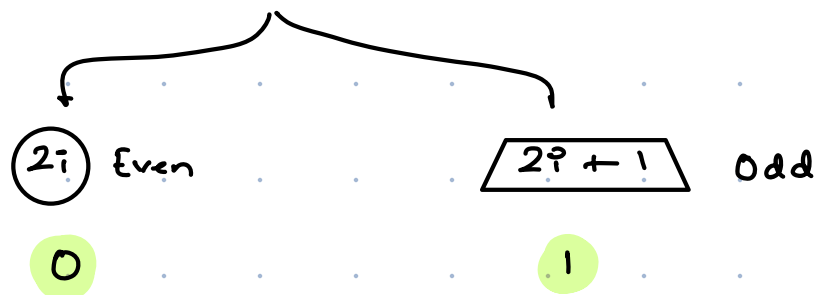
$$cid_x = i \rightarrow pid_x = i/2$$

Obs 2

parent value = 1



parent value = 0



Conclusion → Even idx will always have same value as parent

```
int fun (int n, int k.)
```

```
    if (n == 0 || k == 0) return 0;
```

```
    int p = fun (n-1, k/2);
```

```
    if (k % 2 == 0) return p;
```

```
    else return 1-p;
```

TC: $O(\log k)$

SC: $O(\log k)$

Letter Phone

Problem Description

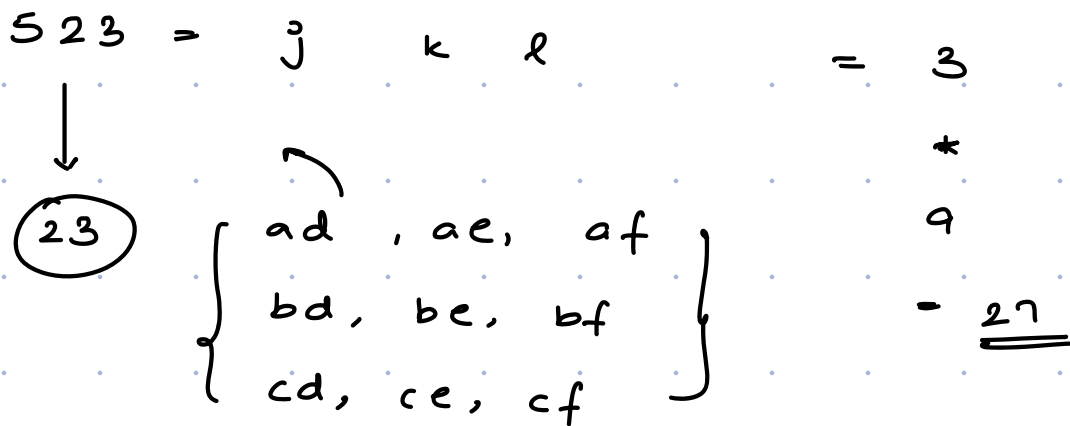
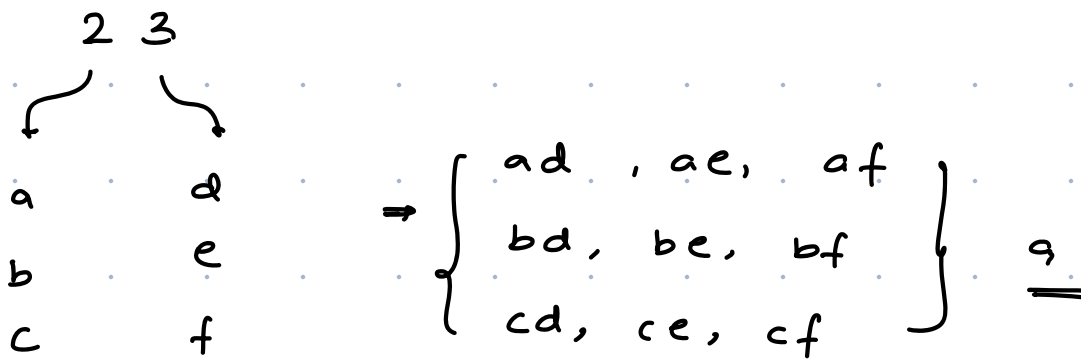
Given a digit string **A**, return all possible letter combinations that the number could represent.

A mapping of digit to letters (just like on the telephone buttons) is given below.



The digit 0 maps to 0 itself. The digit 1 maps to 1 itself.

NOTE: Make sure the returned strings are lexicographically sorted.



```

1 class Solution {
2     public List<String> letterCombinations(String str) {
3         if(str.length()==0) return new ArrayList<>();
4         String[]code={"abc","def","ghi","jkl","mno","pqrs","tuv","wxyz"};
5         return getKPC(str,code);
6     }
7     public ArrayList<String> getKPC(String str,String[]code) {
8         if(str.length()==0){
9             ArrayList<String>base=new ArrayList<>();
10            base.add("");
11            return base;
12        }
13        char ch=str.charAt(0);
14        ArrayList<String>recans=getKPC(str.substring(1),code);
15        int c=ch-'2';
16        ArrayList<String>myans=new ArrayList<>();
17        String codes=code[c];
18        for(int i=0;i<codes.length();i++){
19            char p=codes.charAt(i);
20            for(String a:recans){
21                myans.add(p+a);
22            }
23        }
24        return myans;
25    }
26 }

```

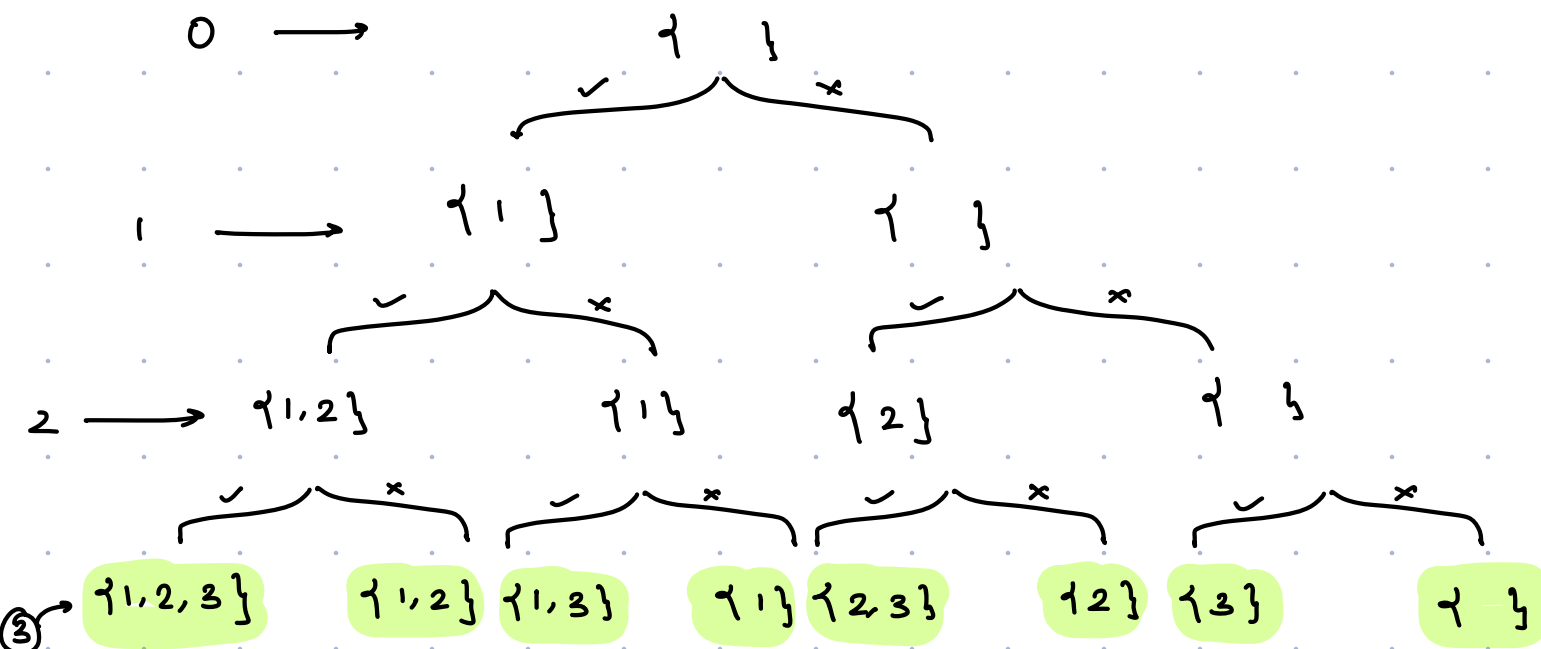
Reset

Test

Test with C

$A[] = \{ \underset{0}{1}, \underset{1}{2}, \underset{2}{3} \} =$

All subsets



{ - - - }

{ - - 3 }

{ - 2 3 }

{ 1, 2, 3 } = { - 2 - }

{ 1 - - }

{ 1 - 3 }

{ 1 2 - }

{ 1 2 3 }

idx, ans }

✓ subset (idx + 1 , ans + A[idx])

x subset (idx + 1 , ans)